Q10) WAP to create Student array Object to hold multiple Student object dynamically.

```
package com.rari.basic;

import java.util.InputMismatchException;
import java.util.Scanner;

//we# to create student array Object to hold multiple Student object dynamically.

record Student(Integer id, String name)
{
}

public class Array2014
{
  public static void main(String[] args)
  {
    Scanner sc = new Scanner(System.in);
    try{
      System.out.print("Enter the number of the Student :");
      int size = Integer.parseInt(sc.next(line));

      Student student[] = new Student[size];

      //Initializing the Student Array
      for(int i=0; i<size; i++)
      {
        System.out.print("Enter the Student Id :");
        int id = Integer.parseInt(sc.next(line));

        System.out.print("Enter the Student Name :");
        String name = sc.next(line);

        student[i] = new Student(id, name);
      }

      System.out.print("Printing the Student Data :");
      for(Student student : student)
      {
        System.out.print(student);
      }
    }
    catch(NegativeArraySizeException e)
    {
      System.out.print("Array size must be positive integer");
    }
    catch(InputMismatchException e)
    {
      System.out.print("Input is invalid");
    }
  }
}
```

Q11) Object array by using Literal.(instanceof)

```
package com.rari.basic;

import java.util.Arrays;

//Object array by using literal.(instance of)

public class Array2015
{
  public static void main(String[] args)
  {
    Object []obj = {11, 90.05, 'A', true, "RITT"};

    System.out.print("Printing Heterogeneous Types of data :");
    System.out.print(Arrays.toString(obj));

    System.out.print("\n----------------------------:");

    //Adding all the integer value from the Object array

    Object []arr = {11, 90.05, 'A', true, "RITT", 14, 100};

    int sum = 0;

    for(Object o : arr)
    {
      if(o instanceof Integer)
      {
        sum = sum + (int) o;
      }
    }

    System.out.print("Sum of all the integers :"+sum);
  }
}
```

Q12) Object array by using new keyword using BLC classes.[Trainer, Manager]

```
package com.rari.basic;

//Object array by using new keyword using BLC classes.[Trainer, Manager]

record Manager(Integer id, String name)
{
}

{

}

record Trainer(String name , int experience)
{
}

{

}

record SalesPun(Integer id, String name, Double totalSale)
{
}

{

}

public class Array2e12
{
  public static void main(String[] args)
  {
    Object []arr = new Object[12];
    arr[0] = new Trainer("Sachin", 25);
    arr[1] = new Manager("Dhoni", 18);
    arr[2] = new SalesPun(20, "Adam", 450000);

    System.out.print("Printing Heterogeneous Objects :");
    for(Object obj : arr)
    {
      System.out.print(arr[0]);
    }
  }
}
```

//Accessing the properties

Q13) Create Book Object to access Book class properties through Book array index.

```
package com.rari.basic;

//Create Book Object to access Book class properties through Book array index.

class Book
{
  String ISBN;
  String author;

}

public class Array2e13
{
  public static void main(String[] args)
  {
    Book book[] = new Book[3];
    book[0] = new Book();
    book[1] = new Book();
    book[2] = new Book();

    //Initializing the Book class properties
    book[0].author = "James";
    book[0].ISBN = "Java";

    book[1].author = "Denis";
    book[1].ISBN = "C Language";

    book[2].author = "Scott";
    book[2].ISBN = "Web Java";

    int index = 0;

    while(index<book.length)
    {
      System.out.print(book[index].ISBN);
      System.out.print(book[index]);
      System.out.print(book[index].author);
      index++;
    }
  }
}
```

Q14) WAP to hold 3 player array Object in an array, search the player based on the id using Scanner class.

```
package com.rari.basic;

import java.util.InputMismatchException;
import java.util.Scanner;

//we# to hold 3 player array Object in an array, search the player
//based on the id using Scanner class.

record Player(Integer id, String name, Double basePrice)
{
}

{

}

public class Array2e14
{
  public static void main(String[] args)
  {
    Player []players = new Player[3];
    players[0] = new Player(45, "Rohit", 25000);
    players[1] = new Player(50, "Virat", 35000);
    players[2] = new Player(55, "Dhoni", 45000);
    players[3] = new Player(60, "Kohli", 55000);
    players[4] = new Player(70, "Dhawan", 45000);

    Scanner sc = new Scanner(System.in);
    try{
      System.out.print("Enter Player Id :");
      int id = Integer.parseInt(sc.next(line));

      for(Player player : players)
      {
        if(player.id() == id)
        {
          System.out.print("Player found with "+id+" "+player);
        }
      }
    }
    catch(InputMismatchException e)
    {
      System.out.print(0);
    }
  }
}
```

Q15) WAP to hold 5 Course array Object in an array, Write a logic to find any duplicate course in available by using id.

```
package com.rari.basic;

//we# to hold 5 Course array Object in an array, write a logic
//to find any duplicate course in available by using id.

record Course(Integer id, String name, Double price)
{
}

{

}

public class Array2e15
{
  public static void main(String[] args)
  {
    Course courses[] = new Course[5];

    courses[0] = new Course(1, "Java", 15000);
    courses[1] = new Course(2, "DTM", 12000);
    courses[2] = new Course(3, "Python", 18000);
    courses[3] = new Course(4, "SQL", 10000);
    courses[4] = new Course(5, "Spring", 20000);
    courses[5] = new Course(1, "C Lang", 10000);

    boolean isDuplicate = false;

    for(int i=0; i<courses.length; i++)
    {
      for(int j=i+1;j<courses.length; j++)
      {
        if(courses[i].id() == courses[j].id())
        {
          System.out.print("Duplicate course found with id :"+courses[i].id());
          isDuplicate = true;
        }
      }
    }

    if(!isDuplicate)
    {
      System.out.print("Duplicate Course Not Found!");
    }
  }
}
```

How to sort the custom object by using Array :

* As we know we can sort our array elements by using **Arrays.sort()** method.
* This sort() method internally using compareTo() or method of java.lang.Comparable<T> interface
* We can say any custom object as a Comparable Object, if the corresponding class implements Comparable<T> interface.
* All the Wrapper classes, String, StringBuffer and StringBuilder implements Comparable<T> interface so all these Objects are Comparable Objects.

**** Difference between Comparable<T> and Comparator<T>